

LECTURE SHEET

EXERCISE-I

(Laws of chemical combination and Mole concept)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The law of conservation of mass holds good for all of the following, except
 - 1) Chemical reactions
 - 2) Nuclear reactions
 - 3) Endothermic reactions
 - 4) Exothermic reactions
- Percentage of copper and oxygen in samples of CuO obtained by different methods were found to be the same. This proves the law of
 - 1) Constant proportions
 - 2) Reciprocal proportions
 - 3) Multiple proportions
 - 4) Conservation of mass
- In compound A, 1.00 g nitrogen combines with 0.57 g oxygen. In compound B, 2.00g nitrogen combines with 2.24 g oxygen. In compound C, 3.00 g nitrogen combines with 5.11 g oxygen. These results obey the following law.
 - 1) Law of constant proportion
 - 2) Law of multiple proportion
 - 3) Law of reciprocal proportion
 - 4) Dalton's law of partial pressure
- In the reaction $\text{Hydrogen}_{(g)} + \text{Oxygen}_{(g)} \rightarrow \text{water vapour}$, the ratio of volumes is 2:1:2. This illustrates the law of
 - 1) conservation of mass
 - 2) combining weights
 - 3) combining volumes
 - 4) all the above

5. LIST-I

- Law of conservation of mass
- Avogadro's Law
- Gay-Lussac's Law of combining volumes
- Law of conservation of energy

LIST-II

- $V_1/V_2 = n_1/n_2$
- $2\text{H}_{2(s)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(s)}$
- 12g of C + 32g of $\text{O}_2 = 44\text{g CO}_2$
- $\text{H}_{2(g)} + \text{Cl}_{2(g)} \rightarrow 2\text{HCl}_{(g)}$
- $\text{H}_{2(g)} + \text{Cl}_{2(g)} \rightarrow 2\text{HCl}_{(g)}$, $\Delta H = -184.6\text{kJ}$

The correct match is

- | | A | B | C | D |
|----|---|---|---|---|
| 1) | 3 | 1 | 4 | 5 |
| 3) | 3 | 1 | 2 | 5 |
-
- | | A | B | C | D |
|----|---|---|---|---|
| 2) | 3 | 1 | 5 | 4 |
| 4) | 1 | 2 | 4 | 5 |
-
- One gram molecule of oxygen is
 - 1) 16 gms of oxygen
 - 2) 32 gms of oxygen
 - 3) 8gms of oxygen
 - 4) 1gm of oxygen
 - 1 gram - atom of oxygen is
 - 1) 1 g of oxygen
 - 2) 16g of oxygen
 - 3) 22.4 g of oxygen
 - 4) 8g of oxygen
 - Which of the following represents the absolute mass of 'Na' atom?
 - 1) 23 gms
 - 2) 23 amu
 - 3) $23 \times 1.66 \times 10^{-24}$ gms
 - 4) $23 \times 1.66 \times 10^{-24}$ amu

9. 1 gram of hydrogen contains 6×10^{23} atoms. Then 4 grams of He contains
1) 6×10^{23} atoms 2) 12×10^{23} atoms 3) 24×10^{23} atoms 4) 1.5×10^{23} atoms
10. The molar mass of a substance is 20g. The molecular mass of the substance is
1) 20g 2) 20 a.m.u 3) 10g 4) 10 a.m.u
11. The number of molecules in one litre of water is (density of water = 1g/mL)
1) $6 \times 10^{23} / 22.4$ 2) 3.33×10^{25} 3) 3.33×10^{23} 4) 3.33×10^{24}
12. Which of the following contains the maximum number of atoms?
1) 10g of CaCO_3 2) 4g of hydrogen 3) 9g of NH_4NO_3 4) 1.8g of $\text{C}_6\text{H}_{12}\text{O}_6$
13. The mixture containing the same number of molecules as that of 14 g of CO is
1) 14g of nitrogen + 16g of oxygen 2) 7g of nitrogen + 16g of oxygen
3) 14g of nitrogen + 8g of oxygen 4) 7g of nitrogen + 8g of oxygen
14. Which of the following is heaviest?
1) 50g of iron 2) 5 moles of nitrogen
3) 0.1 gram atom of silver 4) 10^{23} atoms of carbon
15. If we consider that $1/6$, in place of $1/12$, mass of carbon atom is taken to be relative atomic mass unit, the mass of one mole of a substance will
1) increase two fold
2) decrease twice
3) be a function of molecular mass of the substance
4) remain unchanged
16. If the relative atomic mass of oxygen is 64 units, the molecular mass of CO becomes
1) 112 2) 128 3) 28 4) 7
17. 1 mole of water vapour is condensed to liquid at 25°C . Now this water contains
i) 3 moles of atoms ii) 1 mole of hydrogen molecules
iii) 10 moles of electrons iv) 16 g of oxygen
The correct combination is
1) (i) & (ii) are correct 2) (i) & (iii) are correct
3) (i) & (iv) are correct 4) All are correct
18. The ratio between the number of molecules in equal masses of CH_4 and SO_2 is
1) 1:1 2) 4:1 3) 1:4 4) 2:1
19. 7g of nitrogen occupies a volume of 5 litres under certain conditions. Under the same conditions one mole of a gas, having molecular weight 56, occupies a volume of
1) 40L 2) 20L 3) 10L 4) 80L
20. A mixture of 7g of nitrogen and 8g of oxygen at STP occupies a volume of
1) 11,200 mL 2) 22,400 mL 3) 2240 mL 4) 5600 mL
21. Which of the following has highest mass?
1) 1 gram atom of Iron 2) 5 moles of N_2 3) 10^{24} carbon atoms 4) 44.8 lit of He at STP
22. Which contains more number of molecules?
1) 1 mole of carbon dioxide 2) 4g of hydrogen
3) 33.6 litres of oxygen at STP 4) 6g of helium

23. Assertion (A) : 1 c.c. of Nitrogen at STP contains 2.69×10^{19} molecules.
Reason (R) : Molar volume of an ideal gas at STP contains Avogadro number of molecules.
24. There are as many notes as number of oxygen atoms in 24.8 gm $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5 \text{H}_2\text{O}$ (Mw = 248). A note counting machine counts 48 million notes per day. Number of days taken to count these notes is
1) 10^{17} 2) 10^{16} 3) 10^{10} 4) 10^{12}
25. How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms ?
1) 0.02 2) 3.125×10^{-2} 3) 1.25×10^{-2} 4) 2.5×10^{-2}
26. The number of oxygen atoms present in 50g of calcium carbonate is
1) 6.023×10^{23} 2) 30.1×10^{23} 3) 9.035×10^{23} 4) 1.206×10^{24}
27. Total number of sulphate ions present in 3.92g of chromic sulphate is (Cr = 52, S = 32, O = 16)
1) 1.8×10^{22} 2) 1.8×10^{23} 3) 1.2×10^{21} 4) 6×10^{23}
28. The elements 'A' and 'B' combine together to give two compounds A_2B_3 and AB_2 . The weight of 0.2 mole of A_2B_3 is 26gm. The weight of 0.3 mole of AB_2 is 24gm. Then the atomic weights of A and B respectively
1) 15,20 2) 20,25 3) 20,30 4) 25,30

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The mass of 1.5×10^{20} atoms of an element is 15mg. The atomic mass of the element is
a) 60g b) 60mg c) 60 d) 6
2. Bell metal contains 80% copper. The mass of Bell metal which contains 1.5×10^{20} atoms of copper is (Cu=64)
a) 2mg b) 20mg c) 40mg d) 12.8mg
3. Assuming that air at STP contained 80% by volume of nitrogen, the volume of air at STP that contains 4.8×10^{23} molecules of nitrogen is
a) 18 L b) 44.8 L c) 22.4 L d) 11.2 L
4. Ordinary water contains one part of heavy water per 6000 parts of water by weight. The number of heavy water molecules present in a drop of water of volume 0.01mL is (density of water is 1g/mL)
a) 2.5×10^{16} b) 5×10^{17} c) 5×10^{16} d) 7.5×10^{16}
5. The weight of 600 ml of a mixture of ozone and oxygen is 1 gm at STP. The volume of oxygen in the mixture is
a) 200 ml b) 500 ml c) 400 ml d) 300 ml

More than One correct answer Type Questions

6. Which of the following can explain law of reciprocal proportions?
a) $\text{H}_2\text{O}, \text{CH}_4, \text{CO}_2$ b) $\text{CO}, \text{CO}_2, \text{C}_3\text{O}_2$
c) $\text{HF}, \text{CF}_4, \text{CH}_4$ d) $\text{N}_2\text{O}, \text{NO}_2, \text{N}_2\text{O}_4$
7. 0.5 mole of P_4O_{10} contains
a) 80 gm oxygen b) 2 gram atoms phosphorous
c) 5 gram atoms oxygen d) 10 gram atoms oxygen

8. 3200gm sulphur is present in
 a) 9800 gm H_2SO_4 b) 20 mole H_2SO_4 c) 100 moles H_2SO_4 d) 6400 g SO_2
9. 6.023×10^{22} atoms of Hydrogen can make
 a) 0.05 moles of H_2 molecules b) 0.1 gms of Hydrogen atoms
 c) 0.1 gram-molecules of Hydrogen d) 0.1 gram atoms of Hydrogen
10. Equal volumes of oxygen and ozone at a given temperature and pressure contain equal
 a) number of moles b) masses
 c) number of gramatoms d) number of respective molecules
11. 1 mole of $^{14}_7\text{N}^{-3}$ ions contains
 a) $7 \times 6.023 \times 10^{23}$ electrons b) $7 \times 6.023 \times 10^{23}$ protons
 c) $7 \times 6.023 \times 10^{23}$ neutrons d) $14 \times 6.023 \times 10^{23}$ protons

Linked Comprehension Type Questions

Passage-I :

Isotopes are the atoms of same element; they have same atomic number but different mass numbers. Isotopes have different number of neutrons in their nucleus. If an element exists in two isotopes having atomic masses 'a' and 'b' in the ratio m : n, then average atomic mass will be $\frac{m \times a + n \times b}{m + n}$. Different isotopes of same element have same position in the periodic table. The elements which have single isotope are called monoisotopic elements. Greater is the percentage composition of an isotope, more will be its abundance in nature.

12. The isotopes of chlorine with mass number 35 and 37 exist in the ratio (if its average atomic mass is 35.5)
 a) 1 : 1 b) 2 : 1 c) 3 : 1 d) 3 : 2
13. Atomic mass of boron is 10.81. It has two isotopes namely $^{11}_5\text{B}$ and ^x_5B with their relative abundance of 80% and 20% respectively. The value of x is
 a) 10.05 b) 10 c) 10.01 d) 10.02

Passage-II :

When we concept that one mole of a substance contains the same numbers of elementary entities as one mole of any other substance, we don't actually need to know what that number is. Sometimes however we will need to work with the actual number of elementary entities in a mole of substance. This number is called Avogadro's number.

$$N_A = 6.022137 \times 10^{23} \text{ molecules}$$

The unit mol^{-1} which we read as per mole signifies that a collection of N_A molecular level entities is equivalent to one mole at the macroscopic level. For example a mole of carbon contains 6.02×10^{23} atoms of C. A mole of oxygen gas contains 6.02×10^{23} molecules of O_2 .

14. The numbers of atoms present in 8 g of ozone is
 a) N_A b) $3N_A$ c) $\frac{N_A}{6}$ d) $\frac{N_A}{2}$
15. Which of the following is a reasonable value for the number of atoms in 1.00 g of helium ?
 a) 0.25 b) 4.0 c) 4.1×10^{-23} d) 1.5×10^{23}
16. The dot at the end of a sentence has a mass of about one microgram. Assuming that the black stuff is carbon. No. of carbon atoms present in it is
 a) 5×10^{16} b) 5×10^{17} c) 8×10^{23} d) 12×10^{23}

Matrix Matching Type Questions**17. Column-I**

- A) One gram molecule of oxygen gas
 B) Gram molar volume of H_2
 C) 44 gm of CO_2
 D) 18 gm water

Column-II

- P) one mole of H_2
 Q) one mole of O_2
 R) 22.4lit at STP
 S) $3N_A$ atoms

Integer Type Questions

18. Number of years it would take to spend Avogadro's number of rupees at the rate of 10 lakh rupees per second is $x \times 10^{10}$. Find the value of x .
19. Rahul Dravid wants to wear 6.023×10^{21} "Ag" atoms in the form of a ring. His 'Silver-Gold-Copper' alloy ring consists of 20% of Silver. The mass of the ring is $0.9x$. What is 'x'?
20. A copper plate of 20 cm x 10 cm is to be plated with silver of 1 mm thickness on both the sides. Number of moles of silver required for the plating is (density of silver = 10.8 g m / cc) ____
21. To a sample of an element A (at.wt.=70) another element B (atomic wt = 120) is to be added as an impurity. For the ratio of the atoms in the mixture to be $1:10^{-9}$, $x \times 10^{-8}$ g of B will be required for 35g of A. Find the value of x .

EXERCISE-II**(% composition, Empirical and molecular formula, Equivalent weight)****LEVEL-I (MAIN)****Straight Objective Type Questions**

- A certain metal sulphide MS_2 is used extensively as a high temperature lubricant. If MS_2 is 40.00% by mass sulphur, atomic mass of M is
 1) 60 2) 96 3) 100 4) 80
- Haemoglobin contains 0.33% iron ($Fe = 56$). The molecular weight of haemoglobin is 68000. The number of iron atoms in one molecule of haemoglobin is
 1) 2 2) 3 3) 4 4) 5
- A compound contains 20% sulphur. The molecular weight of the compound could be
 1) 80 2) 240 3) 400 4) 640
- The percentage of nitrogen in Magnesium nitride is
 1) 14 2) 28 3) 42 4) 56
- The percentage of silica in sodium silicate is approximately (Atomic weight of Si = 28)
 1) 25 2) 40 3) 50 4) 60
- The empirical formula of acetic acid is the same as that of
 1) Sucrose 2) Glucose 3) Oxalic acid 4) Formic acid
- The relative number of atoms of different elements in a compound are as follows. A = 1.33, B = 1 and C = 1.5. The empirical formula of the compound is
 1) $A_2B_2C_3$ 2) ABC 3) $A_8B_6C_9$ 4) $A_3B_3C_4$

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21. The equivalent weight of Iodine in the reaction $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow 2\text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$ is [M = mol. wt]
 1) M 2) M/2 3) M/3 4) 2M
22. $\text{H}_3\text{PO}_4 + 2\text{KOH} \rightarrow \text{K}_2\text{HPO}_4 + 2\text{H}_2\text{O}$. Based on the above reaction equivalent weight of H_3PO_4 is
 1) 196 2) 98 3) 49 4) 32.67
23. The equivalent weight of glucose in the reaction $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$ is [M = mol. wt]
 1) M 2) M/2 3) M/24 4) 2M
24. In acidic medium Dichromate ion oxidises Ferrous ion to Ferric ion. If the gram-molecular weight of potassium dichromate is 294 gm, its equivalent weight is
 1) 294 2) 147 3) 49 4) 24.5
25. Assertion (A) : The equivalent weights of nitric acid and crystalline oxalic acid are same.
 Reason (R) : The basicity is same for both the acids.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
26. The oxide of an element contains 67.67% of oxygen. Equivalent weight of the element is
 1) 2.46 2) 3.82 3) 4.36 4) 4.96
27. A bivalent metal has equivalent weight 12. The molecular weight of its oxide is
 1) 16 2) 32 3) 40 4) 52

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. 0.36g of an organic compound on combustion gave 1.1g of CO_2 and 0.54g of H_2O . The percentages of carbon and Hydrogen in the compound are
 a) 75, 25 b) 60, 40 c) 83.33, 16.67 d) 77.8, 22.2
2. 0.66g of a compound gave 112ml of nitrogen measured at STP in the Dumas method. The percentage of Nitrogen in the compound is
 a) 25 b) 41.5 c) 42.4 d) 21.2
3. 2 moles of N_2H_4 loses 16 moles of electrons is being converted to a new compound x. Assuming that all of the N appears in the new compound, what is the oxidation state of N in x ?
 a) -1 b) -2 c) +2 d) +4

More than One correct answer Type Questions

4. Which pair of species have same percentage of carbon ?
 a) CH_3COOH b) $\text{C}_6\text{H}_{12}\text{O}_6$ c) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ d) $\text{C}_2\text{H}_5\text{OH}$
5. Acetic acid and glucose have same
 a) empirical formula b) weight composition of elements
 c) ratio of masses of individual elements d) number of gramatoms of each element per mole
6. A certain compound has the molecular formula X_4O_6 having 57.2% X. Thus,
 a) atomic mass of X is 32 b) X may contain six valence electrons
 c) X is an electropositive metal d) X can be a non-metal

7. 80% carbon is present in an alkane by weight. The possible conclusions are
- The empirical formula of the compound is CH_3
 - The minimum number of carbons in the molecule is 2
 - The compound has gram atoms of C & H in 4:1 ratio
 - This composition suits to all alkanes
8. Equivalent weight and formula weight of reactants are same in the conversions ____
- $\text{AgNO}_3 \rightarrow \text{Ag metal}$
 - $\text{CuO} \rightarrow \text{Cu}_2\text{O}$
 - $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O}$
 - $\text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{Na}_2\text{S}_4\text{O}_6$
9. Which relations between equivalent weight (E) and Molecular weight (M) of reactant are correct for the given change?
- FeCl_2 into $\text{Fe}(\text{OH})_2$; $E = \frac{M}{2}$
 - $\text{Fe}_2(\text{SO}_4)_3$ into Fe^{+2} ; $E = \frac{M}{2}$
 - FeSO_4 into $\text{Fe}_2(\text{SO}_4)_3$; $E = \frac{M}{2}$
 - K_2MnO_4 into KMnO_4 ; $E = \frac{M}{2}$
10. $\text{K}_4[\text{Fe}(\text{CN})_6]$ is converted into CO_3^{2-} , Fe^{+3} ions and NO_3^- ions. Here ____
- 'N' is reduced
 - 'C' is oxidised
 - Iron is oxidised
 - $\text{Eq.wt} = \frac{\text{Formula.wt}}{61}$
11. Select the correct statement(s)
- Equivalent weight of $\text{Ca}(\text{HC}_2\text{O}_4)_2$ is $M/4$ when it is a reducing agent
 - Equivalent weight of $\text{Ca}(\text{HC}_2\text{O}_4)_2$ is $M/2$ when it behaves as an acid
 - $\text{Ca}(\text{HC}_2\text{O}_4)_2$ can be estimated by $\text{MnO}_4^-/\text{H}^+$
 - $\text{Ca}(\text{HC}_2\text{O}_4)_2$ can be estimated by an acid
12. Consider the following reaction.
 $\text{Br}_2 + \text{OH}^- \rightarrow \text{BrO}_3^- + \text{Br}^- + \text{H}_2\text{O}$ (unbalanced).
 Select the correct statement
- Equivalent weight of Br_2 when it is reduced to Br^- is 80
 - Equivalent weight of Br_2 when it is oxidised to BrO_3^- is 96
 - Net equivalent weight of Br_2 is 96
 - It is a disproportionation reaction
13. Which statements are correct for the reaction, $\text{NH}_4\text{NO}_2 \xrightarrow{\Delta} \text{N}_2 + 2\text{H}_2\text{O}$
- The reaction is disproportionation in nitrogen
 - Equivalent weight of reactant is $\frac{1}{3}$ rd of its formula weight
 - NH_4^+ reduced NO_2^-
 - NO_2^- reduced NH_4^+
14. $4\text{Mg} + 10\text{HNO}_3 \rightarrow 4\text{Mg}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + 3\text{H}_2\text{O}$ In this reaction
- 96 gms magnesium can reduce one mole of HNO_3
 - Equivalent weight of reduced HNO_3 is $\frac{1}{8}$ th of its molecular weight
 - Entire HNO_3 involved in the reaction is reduced
 - HNO_3 is reduced to the best possible extent

Linked Comprehension Type Questions**Passage :**

Empirical formula is the simplest formula of the compound which gives the atomic ratio of various elements present in one molecule of the compound. However, the molecular formula of the compound gives the number of atoms of various elements present in one molecule of the compound.

Molecular formula = (Empirical formula) $\times$ n

$$n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}$$

A compound may have same empirical and molecular formula. Both these formulae are calculated using percentage composition of constituent elements.

15. Which of the following represents the formula of a substance which contains 50% oxygen?
 a) N_2O b) CO_2 c) NO_2 d) CH_3OH
16. An oxide of iodine (I = 127) contains 25.4 g of iodine and 8 g of oxygen. Its formula could be
 a) I_2O_3 b) I_2O c) I_2O_5 d) I_2O_7
17. 10 g of hydrofluoric acid gas occupies 5.6 litres of volume at STP. If the empirical formula of the gas is HF, then its molecular formula in the gaseous state will be
 a) HF b) H_2F_2 c) H_3F_3 d) H_4F_4

Matrix Matching Type Questions18. **Column-I**

- A) $\text{Cu}^{2+} + \text{I}^- \rightarrow \text{Cu}_2\text{I}_2$
 B) $\text{Cu} + \text{Cl}^- \rightarrow \text{CuCl}_2 + \dots$
 C) $\text{MnO}_4^- + \text{H}^+ + \text{e}^- \rightarrow \text{Mn}^{2+} + \dots$
 D) $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}_2\text{O}_3 + \dots$

Column-II

- P) Eq. wt. = M.wt/6
 Q) Eq.wt. = M.wt/5
 R) Eq. wt. = M.wt/1
 S) Eq. wt. = M.wt/2

19. **Column-I**

- A) N_2
 B) CO
 C) $\text{C}_6\text{H}_{12}\text{O}_6$
 D) CH_3COOH

Column-II

- P) 40% carbon by mass
 Q) Empirical formula CH_2O
 R) Vapour density : 14
 S) $14N_A$ electrons in a mole

Integer Type Questions

20. 4.4 gms of a hydrocarbon on combustion produced 6.72 L of CO_2 at STP and sufficient water vapour that can generate 0.4gms of H_2 with excess Na. The hydrocarbon is 'x' times heavier than CO_2 . What is 'x'?
21. What is the $\frac{\text{Mol.wt}}{\text{Eq.wt}}$ ratio of $\text{Fe}_2(\text{SO}_4)_3$ being converted into $\text{Fe}(\text{OH})_3$
22. What is $\frac{\text{Mol.wt}}{\text{Eq.wt}}$ of FeC_2O_4 getting converted into Fe^{+3} and CO_2 ?
23. $\text{Eq.wt} = \frac{\text{Mol.wt}}{x}$, What is 'x' for acetaldehyde converted into acetic acid.

24. What is the equivalent weight of methane during its Combustion?
25. For the following reaction,

$$\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$$
 equivalent mass of $\text{N}_2 = \frac{\text{molar mass of N}_2}{x}$. What is the value of x .
26. The equivalent mass of an element is 4. Its chloride has vapour density 59.25. Then the valency of the element is ____

EXERCISE-III

(Stoichiometric calculations, Limiting reagent, Eudiometry)

LEVEL-I (MAIN)
Straight Objective Type Questions

- The number of moles of CO_2 produced when 3 moles of HCl react with excess of CaCO_3 is
 1) 1 2) 1.5 3) 2 4) 2.5
- The mass of carbondioxide obtained when 2g of pure limestone is calcinated is
 1) 44g 2) 0.22g 3) 0.88g 4) 8.8g
- The weight of oxygen required to completely react with 27g of aluminium is
 1) 8g 2) 16g 3) 32g 4) 24g
- Air contains 20% by volume of oxygen. The volume of air required for the complete combustion of 1L of methane under the same conditions is
 1) 2L 2) 4L 3) 10L 4) 0.4L
- What volume of H_2 at NTP is required to convert 2.8g of N_2 into NH_3 ?
 1) 2240 ml 2) 22400 ml 3) 6.72 lit 4) 224 lit
- 'X' litres of carbonmonoxide is present at STP. It is completely oxidised to CO_2 . The volume of CO_2 formed is 11.207 litres at STP. What is the value of 'X' in litres?
 1) 22.414 2) 11.207 3) 5.6035 4) 44.828
- When a sample of baking soda is strongly ignited in a crucible, it suffered a loss in weight of 3.1g. The mass of baking soda is
 1) 16.8 g 2) 8.4g 3) 11.6g 4) 4.2g
- A metal oxide has the formula X_2O_3 . It can be reduced by hydrogen to give free metal and water. 0.1596g of metal oxide requires 6 mg of hydrogen for complete reduction. The atomic mass of metal in amu is
 1) 15.58 2) 155.8 3) 5.58 4) 55.8
- In the reaction $4\text{A} + 2\text{B} + 3\text{C} \rightarrow \text{A}_4\text{B}_2\text{C}_3$, what will be the number of moles of product formed, starting from one mole of A, 0.6 moles of B and 0.72 moles of C
 1) 0.25 2) 0.3 3) 0.24 4) 2.32
- The number of moles of Fe_2O_3 formed when 5.6 lit of O_2 reacts with 5.6g of Fe is
 1) 0.125 2) 0.01 3) 0.05 4) 0.10
- 1 mole of each of Ca(OH)_2 and H_3PO_4 are allowed to react under dilute conditions. The maximum number of moles of $\text{Ca}_3(\text{PO}_4)_2$ formed is
 1) 1 2) 1/2 3) 1/3 4) 3

12. 10 g of CaCO_3 is completely decomposed to X and CaO. X is passed into an aqueous solution containing one mole of sodium carbonate. What is the number of moles of sodium bicarbonate formed? (Mol. wts: $\text{CaCO}_3 = 100$, $\text{Na}_2\text{CO}_3 = 106$, $\text{NaHCO}_3 = 84$)
 1) 0.2 2) 0.1 3) 0.01 4) 10
13. The volume of carbondioxide formed when 1 litre of O_2 reacts with 2 lit of CO under the same conditions is
 1) 1L 2) 2L 3) 3L 4) 1.5L
14. 20 ml of nitric oxide combines with 10 ml of oxygen at STP to give NO_2 . The final volume will be
 1) 30 ml 2) 20 ml 3) 10 ml 4) 40 ml

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. 1 lit of oxygen and 3 lit of SO_2 at STP are reacted to produce sulphur trioxide. The the ratio between volume of sulphur trioxide and that of sulphur dioxide after reaction and weight of SO_3 formed (in grams) respectively are
 a) 1:2 ; 7.14 gms b) 2:1 ; 14.28gms
 c) 2:1 ; 7.14 gms d) 1:1 ; 14.28 gms
2. The mass of 80% pure H_2SO_4 required to completely neutralise 60g of NaOH is
 a) 92g b) 58.8g c) 73.5g d) 98g
3. 70g of a sample of magnesite on treatment with excess of HCl gave 11.2L of CO_2 at STP. The percentage purity of the sample is
 a) 80 b) 70 c) 60 d) 50
4. The volume in litres of CO_2 liberated at STP, when 10g of 90% pure limestone is heated completely is
 a) 2.016 b) 20.16 c) 2.24 d) 22.4
5. NH_3 is formed in the following steps
 I. $\text{Ca} + 2\text{C} \rightarrow \text{CaC}_2$ 50% yield
 II. $\text{CaC}_2 + \text{N}_2 \rightarrow \text{CaCN}_2 + \text{C}$ 100% yield
 III. $\text{CaCN}_2 + 3\text{H}_2\text{O} \rightarrow 2\text{NH}_3 + \text{CaCO}_3$ 50% yield
 To obtain 2 moles NH_3 , calcium required is
 a) 1 mol b) 2 mol c) 3 mol d) 4 mol
6. 18.4g of a mixture of CaCO_3 and MgCO_3 on heating gives 4.0g of magnesium oxide. The volume of CO_2 produced at STP in this process is
 a) 1.12 L b) 4.48 lit c) 2.24 L d) 3.36 L
7. A mixture of Na_2CO_3 and NaHCO_3 having a total weight of 100 gm on heating produced 11.2L of CO_2 under STP conditions. The percentage of Na_2CO_3 in the mixture is
 a) 55.8% b) 44.2% c) 84% d) 16%
8. 10ml of mixture containing CO and N_2 requires 7 ml of oxygen to form CO_2 and NO respectively on combustion. Find the volume of N_2 in the mixture
 a) 7/2 b) 17/2 c) 4 ml d) 7ml

9. 40 ml gaseous mixture of CO , CH_4 and Ne was exploded with 10 ml of oxygen. On cooling, the gases occupied 36.5 ml. After treatment with KOH the volume reduced by 9 ml and again on treatment with alkaline pyrogallol, the volume further reduced, percentage of CH_4 in the original mixture is
 a) 22.4 b) 77.5 c) 7.5 d) 15
10. 40 ml of a hydrocarbon undergoes combustion in 260ml of oxygen and gives 160ml of CO_2 . If all volumes are measured under similar conditions of temperature and pressure, the formula of the hydrocarbon is
 a) C_3H_8 b) C_4H_8 c) C_6H_{14} d) C_4H_{10}
11. Phosphine on decomposition produces phosphorus and hydrogen. When 100 ml of phosphine is decomposed the change in volume under laboratory conditions is
 a) 50 ml increase b) 50 ml decrease c) 900 ml decrease d) 75 ml increase

More than One correct answer Type Questions

12. Two moles of H_2SO_4 will be neutralised by
 a) 1 mole KOH b) 2 mole Ca(OH)_2 c) 4 mole KOH d) 2 mole KOH
13. In which of the following reactions, no change in gaseous volume occurs when measured at similar T and P?
 a) Combination of N_2 and O_2 to give NO b) Combination of N_2 and H_2 to form NH_3
 c) Combustion of carbon to give CO_2 d) Combustion of carbon monoxide
14. Which of the following requires six times their volume of oxygen for complete combustion?
 a) n-Butane b) 1-Butene c) 2-Butene d) Cyclobutane
15. One mole Barium chloride and one mole Sodium phosphate are mixed in aqueous medium.
 a) BaCl_2 acts as limiting reagent b) Na_3PO_4 acts as limiting reagent
 c) Half mole of $\text{Ba}_3(\text{PO}_4)_2$ is formed d) 0.33 mole of $\text{Ba}_3(\text{PO}_4)_2$ is formed
16. One litre of CO_2 is passed over hot coke. The volume becomes 1.4 lit. The resultant mixture contains
 a) 0.6 lit CO_2 b) 0.6 lit CO c) 0.8 lit CO d) 0.8 lit CO_2
17. If 0.80 mole of MnO_2 and 146 g of HCl react; $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$ then
 a) 0.80 mole of Cl_2 is formed b) 0.80 mole of HCl remains unreacted
 c) MnO_2 is completely reacts d) MnO_2 is the limiting reactant

Linked Comprehension Type Questions

Passage-I :

10 moles of SO_2 and 4 moles of O_2 are mixed in a closed vessel of volume 2 litres. The mixture is heated in the presence of Pt catalyst. Following reaction takes place:
 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$. Assuming the reaction proceeds to completion.

18. Select the correct statement
 a) SO_2 is the limiting reagent b) O_2 is the limiting reagent
 c) Both SO_2 and O_2 are limiting d) Cannot be predicted
19. Number of moles of SO_3 formed in the reaction will be
 a) 10 b) 4 c) 8 d) 14
20. Number of moles of excess reactant remaining
 a) 4 b) 2 c) 6 d) 8

Passage-II :

In a reaction vessel, 100 g H_2 and 100 g Cl_2 are mixed and suitable conditions are provided for the reaction: $H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$

21. The amount of HCl formed in this reaction (at 100% yield) will be
 a) 102.8 g b) 73 g c) 36.5 g d) 142 g
22. The amount of excess reactant remaining is
 a) 50 g b) 97.2 g c) 46 g d) 64 g
23. The amount of HCl formed (at 90% yield) will be
 a) 36.8 g b) 62.5 g c) 80 g d) 92.53 g

Matrix Matching Type Questions

24. **Column-I** **Column-II (Limiting reagent)**

A) N_2 + $3H_2 \longrightarrow 2NH_3$ 0.2 mole 0.7 mole	P) H_2
B) H_2 + $2C \longrightarrow C_2H_2$ 1g 24g	Q) C
C) C + $O_2 \longrightarrow CO$ 0.5g 2.24lit	R) O_2
D) $2H_2$ + $O_2 \longrightarrow 2H_2O$ 2g 2g	S) N_2

25. Match type of the salt formed with the appropriate reaction.

Column-I (Reaction)**Column-II (Type of Salt)**

A) 1 mole of oxalic acid + 1 mole of NaOH	P) Mixed salt
B) 1 mole of H_3PO_2 + 1 mole KOH	Q) Basic salt
C) 1 mole of $Ca(OH)_2$ + 1 mole of HCl	R) Acid salt
D) Dry slaked lime + Cl_2	S) Normal salt

Integer Type Questions

26. Sulphur trioxide is prepared by the following two reactions :
 $S_{8(s)} + 8O_{2(g)} \rightarrow 8SO_{2(g)}$; $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$.
 How many grams of SO_3 are produced from 1.6g S_8 ?
27. 2 moles of pure $KClO_3$ is decomposed to an extent of 66.6%. How many moles of O_2 is released ?
28. Nitric acid can be produced from NH_3 in three step process
 I) $4NH_{3(g)} + 5O_{2(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$
 II) $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$
 III) $3NO_{2(g)} + H_2O_{(l)} \rightarrow 2HNO_{3(aq)} + NO_{(g)}$
 % yield of I, II, III reaction are respectively 50%, 60% and 80%. Then how much volume of $NH_{3(g)}$ at STP is required to produce 2.25 gm of HNO_3 .